

Animated Ink Bleeding with Computational Fluid Dynamics

Grzegorz Gruszczynski
IDEAS NCBR
Poland
grzegorz.gruszczynski@ideas-ncbr.pl

Mateusz Tokarz
PLATIGE IMAGE
Poland
mtokarz@platige.com

Przemysław Musiański
IDEAS NCBR
Poland
New Jersey Institute of Technology
United States
przem@njit.edu

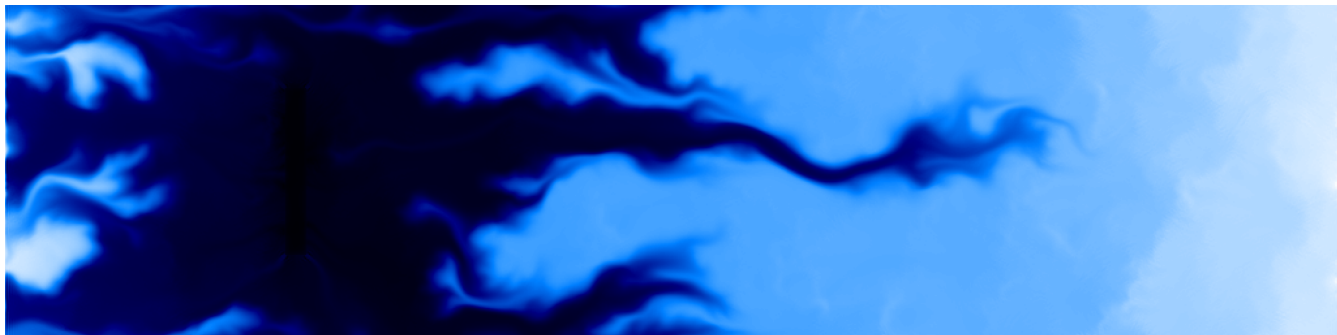


Figure 1: Visualization of the ink flow through a porous paper demonstrating the usage of computational fluid dynamics for artistic effects.

ABSTRACT

This work presents an advanced computational approach for simulating the traditional ink-on-wet paper art technique, focusing on the dynamic interplay between ink, water, and paper using the Lattice Boltzmann Method (LBM). We model ink motion using the advection-diffusion equation coupled with Navier-Stokes equations in a porous medium. We explore the impact of the background material's permeability on the realism of digital ink simulations. Our results highlight this method's potential in digital art, offering artists and animators a novel tool for creating realistic ink effects.

CCS CONCEPTS

• **Computing methodologies** → **Physical simulation**; Modeling methodologies.

KEYWORDS

ink bleeding, watercolor, porous media, computational fluid dynamics, lattice Boltzmann method

ACM Reference Format:

Grzegorz Gruszczynski, Mateusz Tokarz, and Przemysław Musiański. 2024. Animated Ink Bleeding with Computational Fluid Dynamics. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671060>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671060>

1 INTRODUCTION

Traditional watercolor art beautifully captures the fluid interplay between ink and water on paper, creating visually complex scenes. Digital artists, inspired by these traditional methods, have attempted to replicate this effect in visual media.

However, due to the deficiency of dedicated software tools, these digital versions may be cumbersome to produce. We investigate a simulation technique originating from mining engineering [Whitaker 1986]. We show that the flow branching effect can be achieved by introducing an appropriate forcing term to the Navier-Stokes equations. Moreover, the effective viscosity is locally modified to account for additional viscous shear stress experienced by the fluid flowing through a porous medium.

Although our work focuses on the modeling of ink flow, it can be easily extended by incorporating additional effects related to various aspects of the watercolor painting process. Pigment adsorption and desorption have been detailed in [Curtis et al. 1997]. The paper drying effect was studied in [Chu and Tai 2005; Curtis et al. 1997; Ďurikovič and Páleníková 2017]. The explicit representation of pigment particles was explored by [Ďurikovič and Páleníková 2017]. Finally, pigment colors can be layered using Kubelka and Munks' theory [Kubelka and Munk 1931].

2 TECHNICAL SOLUTION

We perform the simulations using a numerical scheme based on the Lattice Boltzmann Method (LBM) implemented in the TCLB framework [Łaniewski Wołk and Rokicki 2016]. Unlike traditional methods that solve the Navier-Stokes equations directly, LBM operates on a mesoscopic scale. At this scale, the probability distribution function is used to specify the spatial density of macroscopic quantities like mass or chemical concentration.

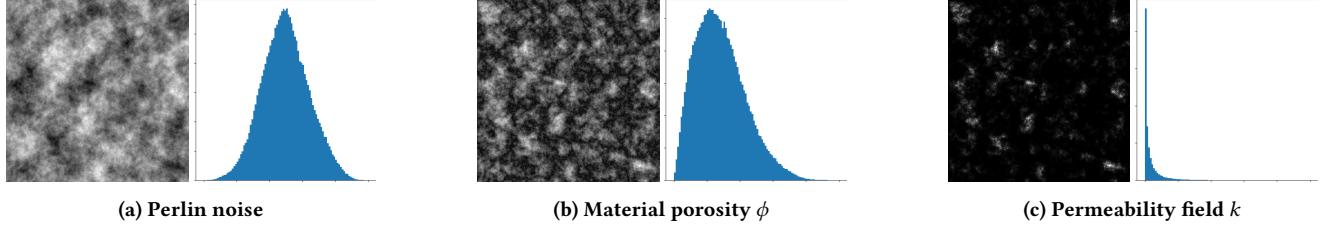


Figure 2: Structure of the material is generated with Perlin noise. The porosity, ϕ , is modeled as the χ -distribution of the Perlin noise, which visually reassembles the small-scale, high-frequency details. The final permeability field, k , is obtained from the porosity using the Carman-Kozeny equation. The histograms of the distributions are shown on the right side, respectively.

In the discretized approach, the discrete distributions are transmitted to adjacent nodes via streaming, involving data exchange among nodes. Further, each node undergoes a local collision step that nudges the distribution function toward equilibrium. Due to the locality of the collision step, efficient parallel computations of the LBM are possible, making it well-suited for GPU implementations. Moreover, the Cartesian grid naturally fits the pixels building the image. Our model uses a two-distribution approach to distinguish the hydrodynamic field, governed by the Navier-Stokes equations, and the advected passive scalar (ink) on the D2Q9 lattice.

We solve the full Navier-Stokes equations,

$$\begin{cases} \frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = 0 \\ \rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot (\mu^{\text{eff}} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]) + \mathbf{F}_D, \end{cases}$$

where the fluid's density, velocity, and pressure are subsequently denoted with the ρ , $\mathbf{u}$ and p symbols. The effective dynamic viscosity of the fluid, $\mu^{\text{eff}} = \mu + \mu^B$, is composed of the fluid viscosity, μ , and an additional term, μ^B , due to the presence of the porous medium. We use the Brinkman model to find μ^B , following the formula proposed in [Ginzburg et al. 2015]. The spatially variable flow resistance, $\mathbf{F}_D$, related to the permeability field is calculated via the Darcy law, $\mathbf{F}_D = -\mathbf{u} \frac{\mu^{\text{eff}}}{k}$. The motion of the ink is simulated using the advection-diffusion equation,

$$\frac{\partial \rho I}{\partial t} + \nabla \cdot (\rho I \mathbf{u}) = \nabla \cdot D^{\text{eff}} \nabla I + G_D,$$

where I denotes the ink's intensity. The effective diffusivity, D^{eff} , and flux resistance, G_D , are calculated analogously as for the hydrodynamics part. The background material's porosity, ϕ , is generated using the Perlin noise [Perlin 1985]—a cost-effective and versatile model for procedural generation of complex textures (cf. Figure 2a). Next, we construct the $\chi(2)$ distribution from the Perlin noise. This approach turned out to serve as a good approximation of the porosity $\phi = \chi(P, 2)$ of the paper background texture in our experiments (cf. Figure 2b). Subsequently, the permeability field of the material, k , is computed using the Carman-Kozeny equation, $k = \frac{\phi^3}{\beta(1-\phi)^2 + \epsilon}$, where β is a tunable parameter (cf. Figure 2c).

3 RESULTS

Our preliminary results indicate the crucial role of the background porosity in visually appealing effects. The high-frequency features are responsible for flow branching (diverging) through the porous media. This makes the simulations challenging to control as a slight

change in the underlying structure, such as crack or inclusion, may cause a flow bifurcation. This kind of *butterfly effect* can easily mislead commonly used gradient optimization techniques. Figure 1 shows the flow patterns simulated on a 1024x256 D2Q9 lattice using our approach. The computations took ~ 3 minutes using a desktop RTX 2080S GPU. Our work concludes that a realistically looking ink-flow effect can be achieved by combining fluid simulation with an appropriate model of the background material.

4 FUTURE GOALS

We aim to extend our solver to be executed in multiple modes: (1) The forward mode would allow the artist to run an interactive simulation during which the parameters, such as the background porosity, can be adjusted with a brush-like tool. (2) Alternatively, the optimization mode would adjust the background material properties to match the desired flow contours. Here, we plan to use the Taichi framework [Hu et al. 2020] and its high-performance, parallel (GPU-ready), and differentiable code. (3) Finally, we intend to replace the numerical optimizer by designing a generative neural network capable of proposing the material properties.

REFERENCES

- Nelson S.-H. Chu and Chiew-Lan Tai. 2005. MoXi: real-time ink dispersion in absorbent paper. *ACM Trans. Graph.* 24, 3 (jul 2005), 504–511. <https://doi.org/10.1145/1073204.1073221>
- Cassidy J. Curtis, Sean E. Anderson, Joshua E. Seims, Kurt W. Fleischer, and David H. Salesin. 1997. Computer-generated watercolor. In *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '97)*. ACM Press/Addison-Wesley Publishing Co., USA, 421–430. <https://doi.org/10.1145/258734.258896>
- Irina Ginzburg, Goncalo Silva, and Laurent Talon. 2015. Analysis and improvement of Brinkman lattice Boltzmann schemes: Bulk, boundary, interface. Similarity and distinctness with finite elements in heterogeneous porous media. *Phys. Rev. E* 91 (Feb 2015), 023307. Issue 2. <https://doi.org/10.1103/PhysRevE.91.023307>
- Yuanming Hu, Luke Anderson, Tzu-Mao Li, Qi Sun, Nathan Carr, Jonathan Ragan-Kelley, and Frédo Durand. 2020. Diff Taichi: Differentiable Programming for Physical Simulation. *ICLR* (2020).
- P. Kubelka and F. Munk. 1931. Ein Beitrag zur Optik der Farben Striche. *Zürich Tech. Physik* 12 (1931), 593.
- Ken Perlin. 1985. An image synthesizer. In *Proceedings of the 12th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '85)*. Association for Computing Machinery, New York, NY, USA, 287–296. <https://doi.org/10.1145/325334.325247>
- Stephen Whitaker. 1986. Flow in porous media I: A theoretical derivation of Darcy's law. *Transport in porous media* 1 (1986), 3–25.
- Roman Ďuríkovič and Zuzana Páleníková. 2017. Real-Time Watercolor Simulation with Fluid Vorticity Within Brush Stroke. In *2017 21st International Conference Information Visualisation (IV)*. 158–163. <https://doi.org/10.1109/IV.2017.10>
- L. Łaniewski Wolik and J. Rokicki. 2016. Adjoint Lattice Boltzmann for topology optimization on multi-GPU architecture. *Computers and Mathematics with Applications* 71, 3 (2016), 833–848. <https://doi.org/10.1016/j.camwa.2015.12.043>